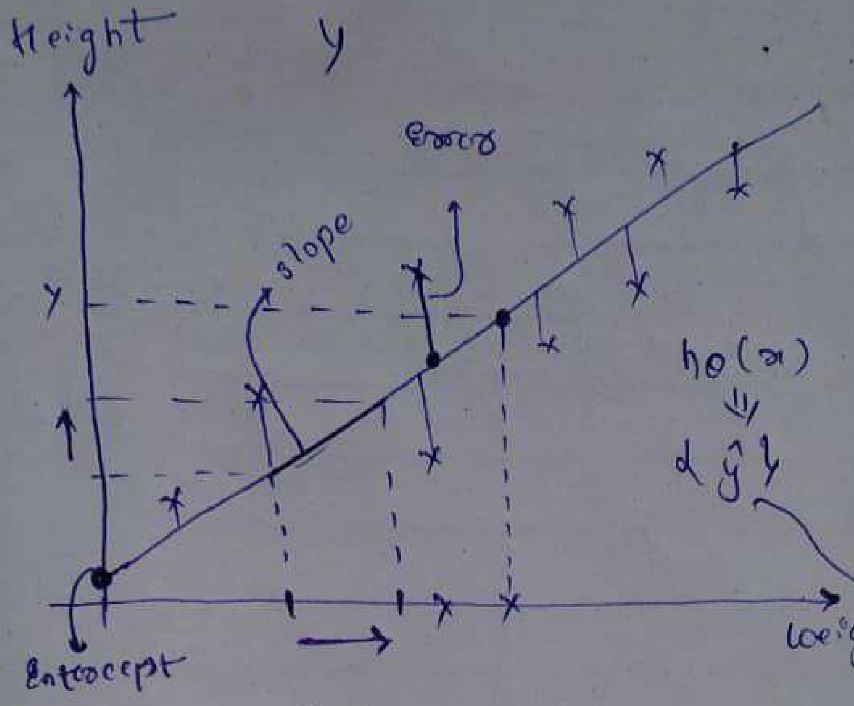
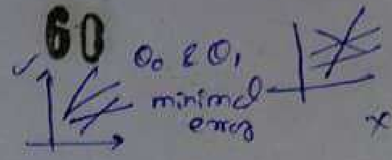
Our main aim is to come with best fit line which has minimal error. (Create a new best fit line which can minimize the error)





Dataset

$y = mx + c$

$y = \beta_0 + \beta_1 x$

$h_\theta(x)$

$\Downarrow$

$\{\hat{y}\}$ → predicted point

$$\boxed{h_\theta(x) = \theta_0 + \theta_1 x}$$

$$h_\theta(x) = \theta_0 + \theta_1 x$$

if $x = 0$

$h_\theta(x) = \theta_0$

θ_0 = Intercept

θ_1 = slope or coefficient

Error $(y - \hat{y})$

- difference between the points

Note: The unit movement in x axis with respect to y axis that indicates something called as slope.

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 \dots\dots\dots + \theta_n x_n$$

Note: There are one independent feature (weight) then make one slope, or if multiple feature make multiple slope.